

Code-Layout, Readability and Reusability

Date	15 July 2026
Team ID	SWTID-2026-3798
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed(Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper Indentation
2	Proper File Structure	Files and folders are logically organized	Yes	Organized folders
3	Meaningful Variable Names	Variables reflect their purpose clearly		Meaningful names
4	Function / Method Names	Functions are descriptively named	Yes	Clear functions
5	Code Comments	Inline and block comments explain log c	Yes	Useful comments
6	Modular Design	Code is split into reusable functions/modules	Yes	Modular code
7	No Redundant Code	Duplicate or unused code is removed	Yes	No duplicate code
8	Error Handling Exceptions and errors are handled gracefully	Exceptions and errors are handled gracefully	Partial	Basic validation

Reusable Components / Modules:

S.No	Component / Module Name	Language/ Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing	Python	Training	High

2	XGBoost Model	Python	Prediction	High
3	StandardScaler	Scikit-learn	Flask App	High
4	Flask App	Flask	Deployment	High
5	HTML/CSS/JS	Frontend	UI	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well Organized
Readability	5	Easy to read
Reusability	5	Reusable modules
Documentation / Comments	4	Good comments
Overall Score	5	Clean & maintainable